

Lesson 5: SLR-ish: Categorical Covariates

Nicky Wakim

2026-01-21

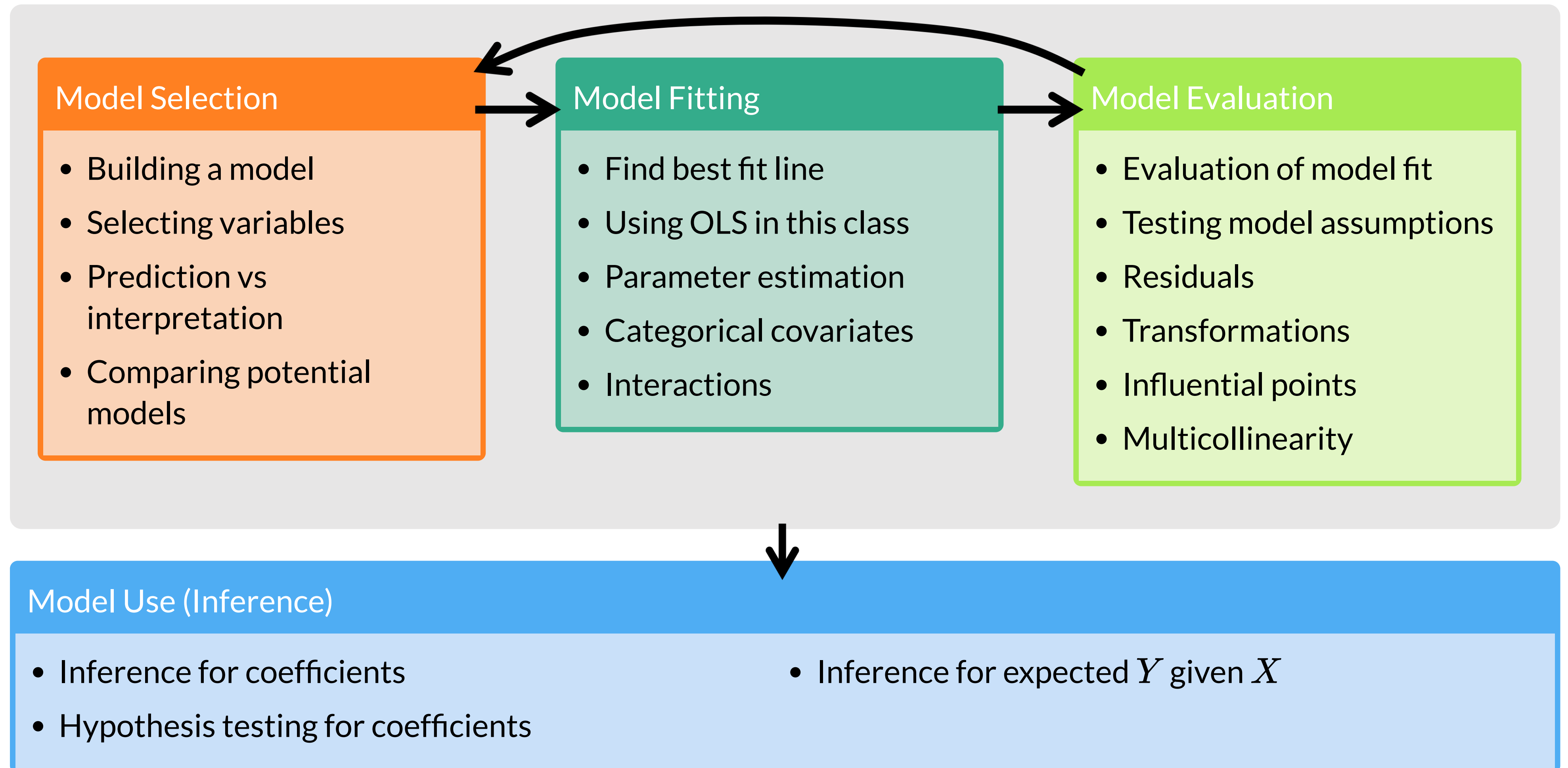
Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables
2. Write the regression equation for a categorical variable using reference cell coding
3. Calculate and interpret coefficients for reference cell coding
4. Change the reference level in a categorical variable for reference cell coding
5. Create new variables and interpret coefficient for ordinal / scoring coding

Why “SLR-*ish*”?

- The **strict** definition of simple linear regression: only two variables that are BOTH continuous
- Common (but kinda wrong) use of simple linear regression: only two variables with outcome continuous and predictor not specified
- I’m including multi-level categorical covariates in SLR mostly because it’s easier to learn now!

Let's map that to our regression analysis process



Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables
2. Write the regression equation for a categorical variable using reference cell coding
3. Calculate and interpret coefficients for reference cell coding
4. Change the reference level in a categorical variable for reference cell coding
5. Create new variables and interpret coefficient for ordinal / scoring coding

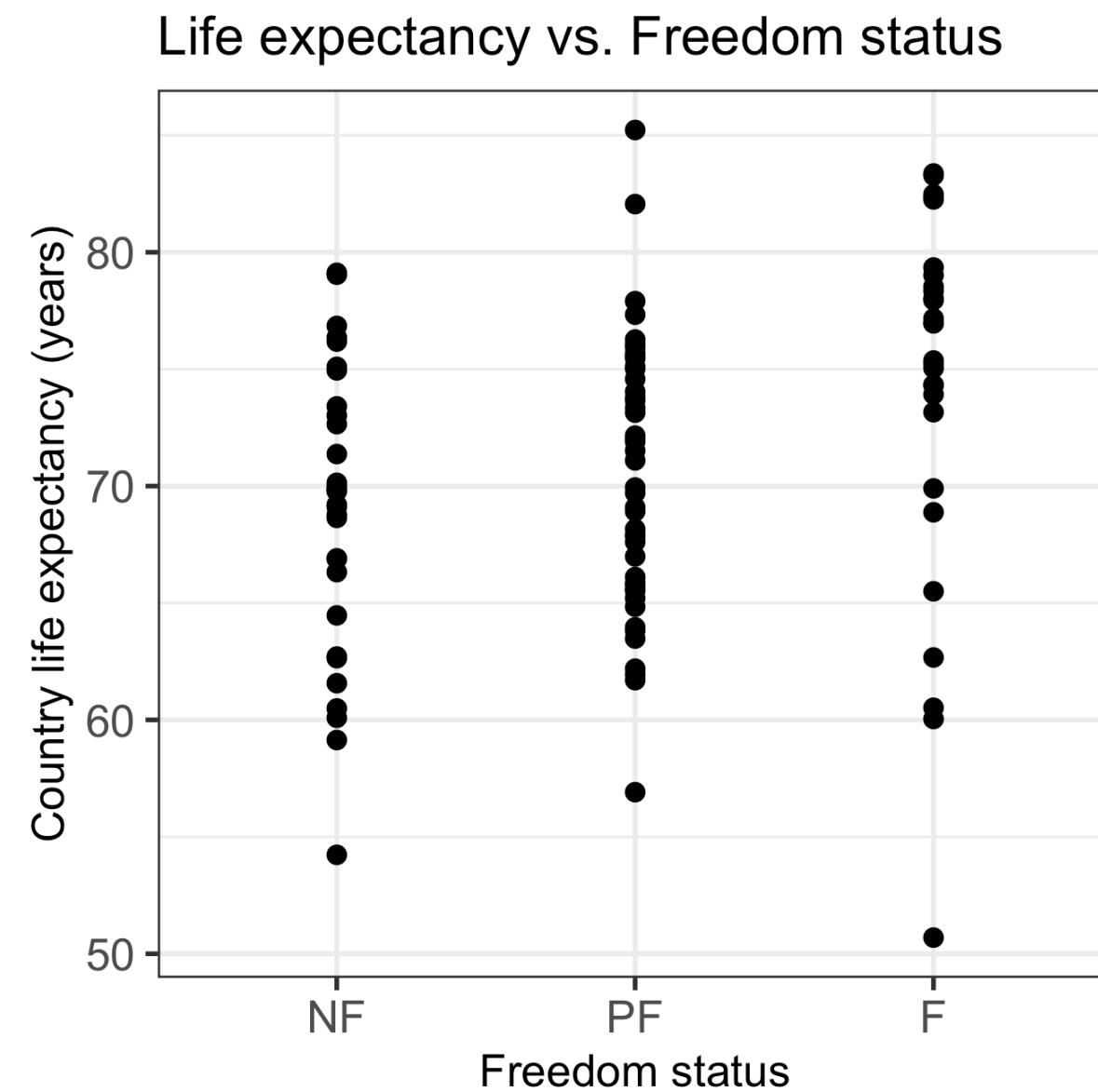
Still looking at Gapminder Life Expectancy data

- We will look at life expectancy vs. these Freedom status
- You can access my codebook [here](#)
- Gapminder measures freedom status: This is a descriptive text of the real-world rights and freedoms enjoyed by individuals. It is determined by the freedom rating which is the average of political rights and civil liberties ratings.
 - Not free (NF)
 - Partly free (PF)
 - Free (F)
- Freedom status is a **multi-level categorical covariate**: it has three statuses
- Note: I am calling the expected life expectancy $\widehat{LE}$
 - Previously, I have written life $\widehat{\text{expectancy}}$

Visualizing the relationship

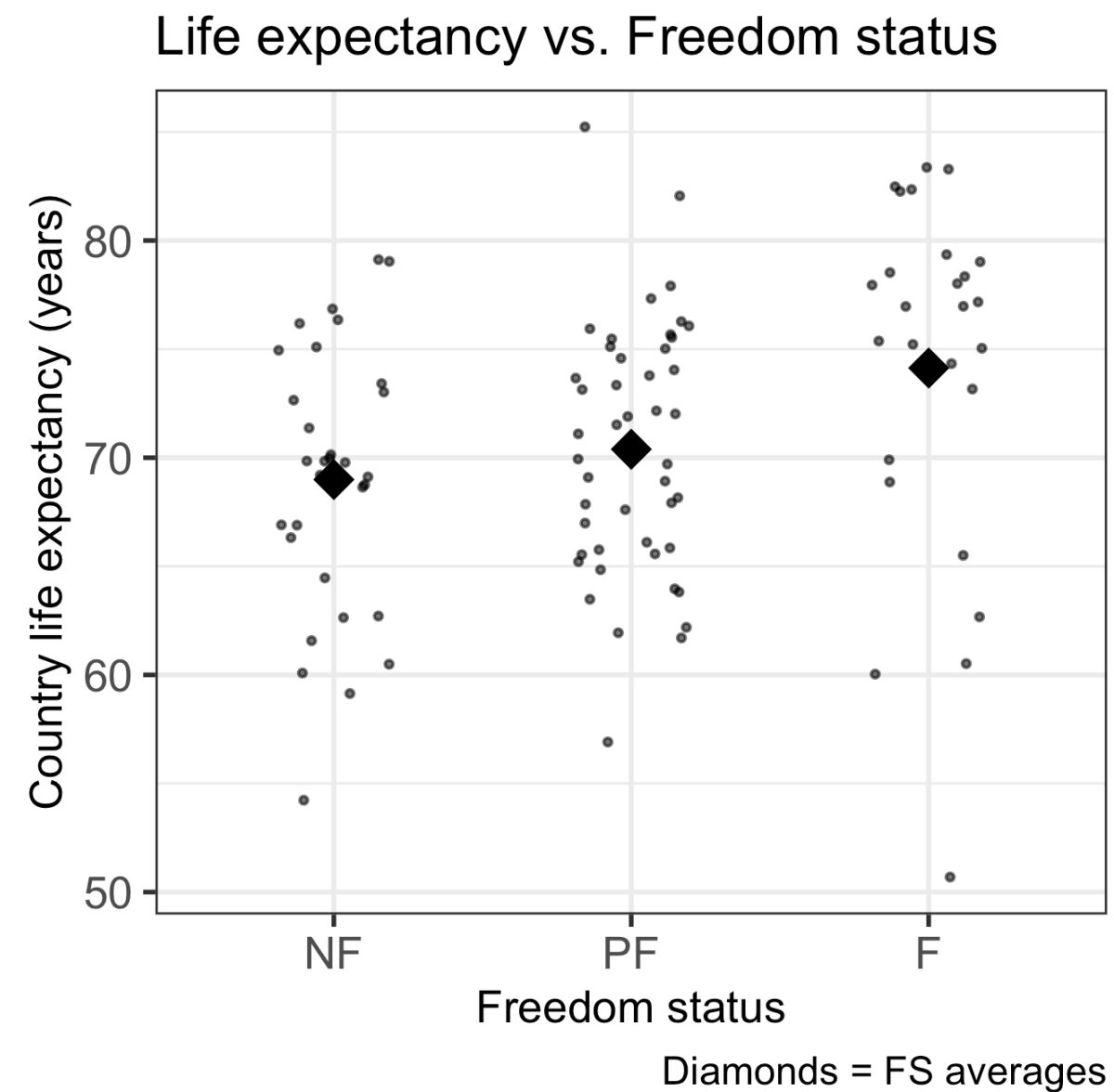
Bad option for visualization:

► [Code](#)



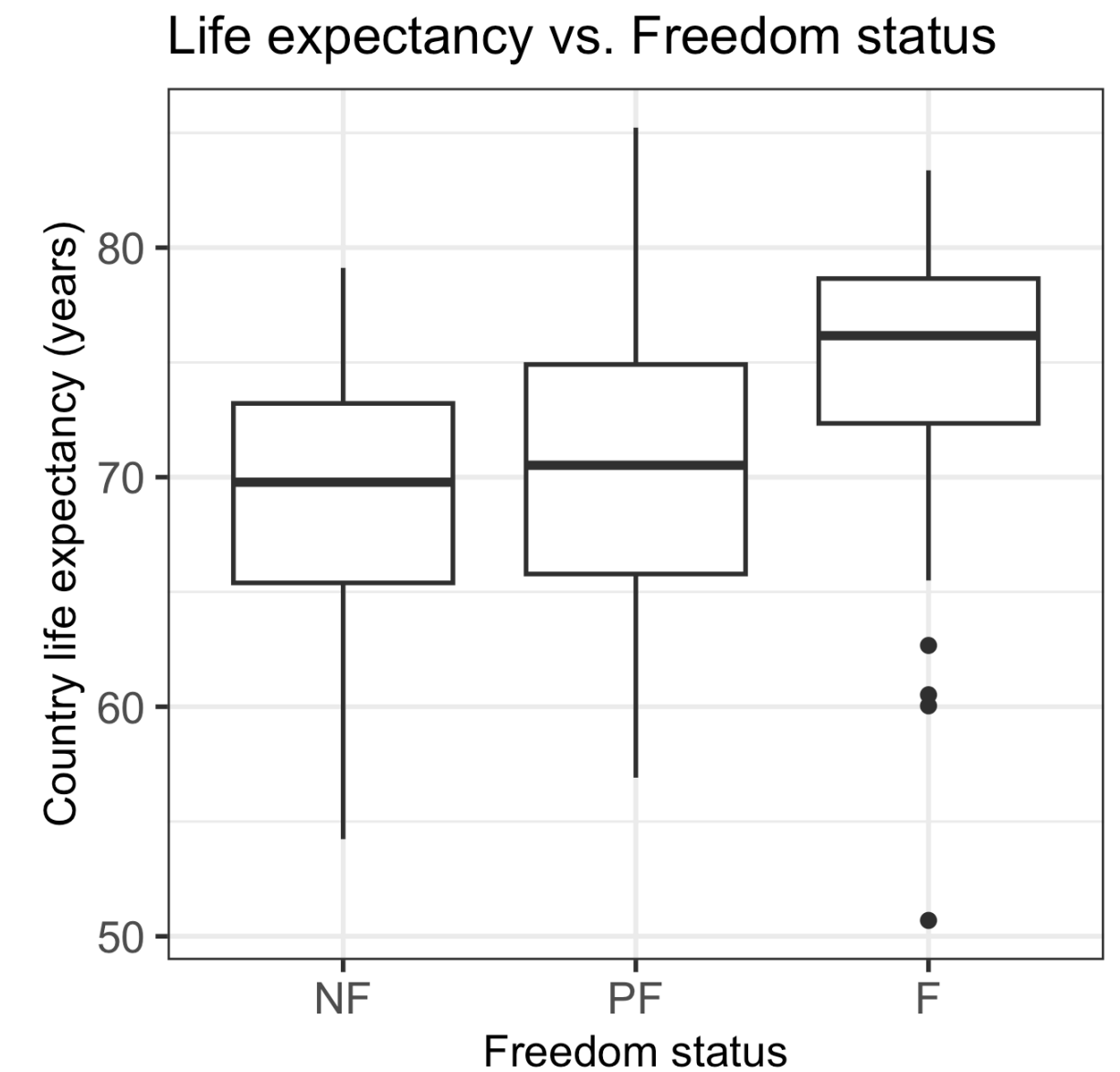
Good option for visualization:

► [Code](#)



Good option for visualization:

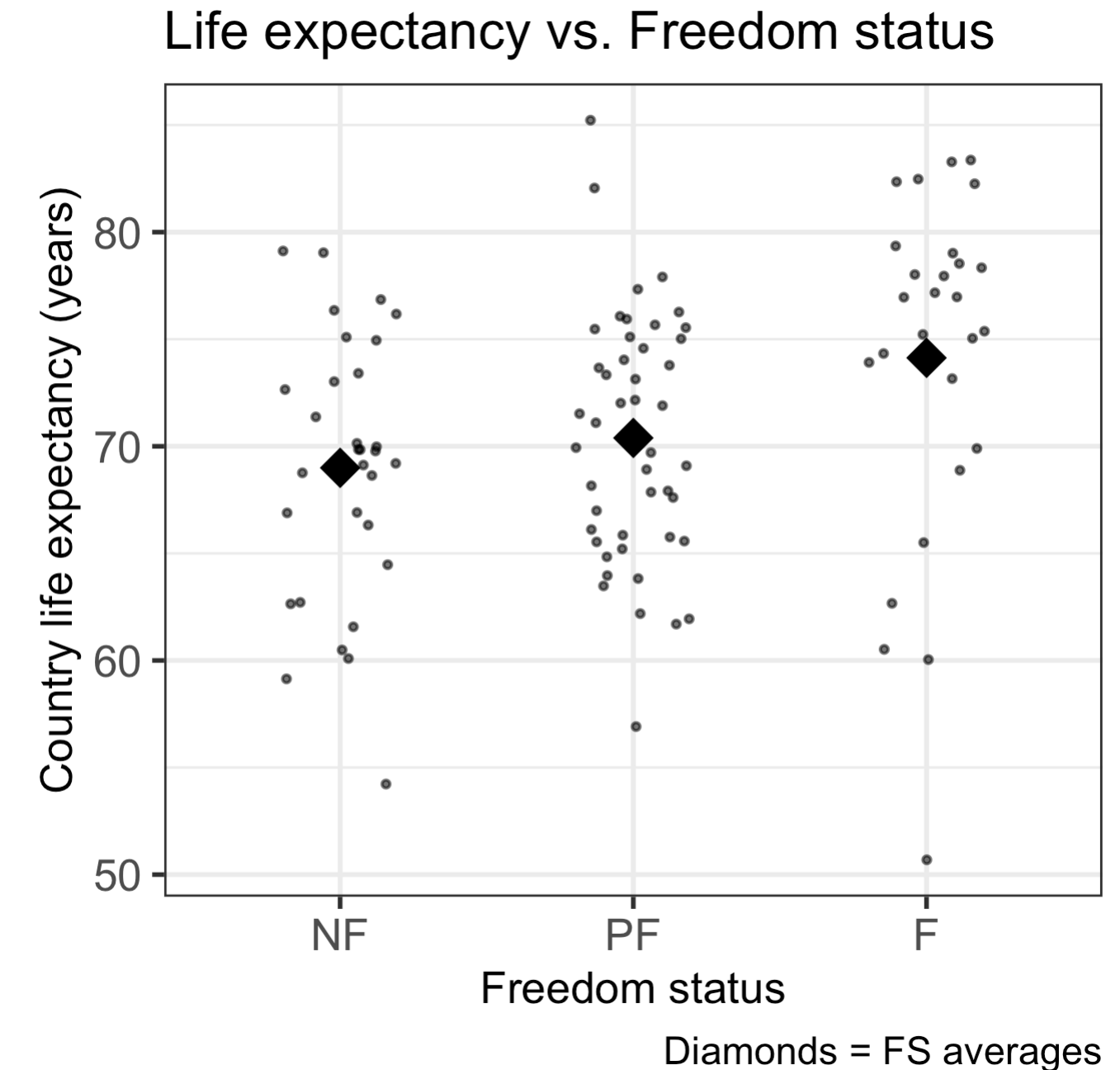
► [Code](#)



- Used `geom_jitter()`

Linear regression with a categorical covariate

- When using a categorical covariate/predictor,
 - We do **NOT**, technically, find a best-fit line
- Instead we model the **means** of the outcome
 - For the different levels of the categorical variable
- In 511/525, we used Kruskal-Wallis test and our ANOVA table to test if groups means were statistically different from one another
- We can do this **using linear models** AND we can include other variables in the model



There are different ways to code categorical variables

Reference Cell Coding

- Sometimes called dummy coding
- Compares each level of a variable to the omitted (reference) level

Effect coding

- Sometimes called sum coding or deviation coding
- Compares deviations from the grand mean
- Not covered in our class

Ordinal encoding

- Sometimes called scoring
- Categories have a natural, even spaced ordering
- Linear relationship between levels

If you want to learn more about these and other coding schemes:

- [Coding Systems for Categorical Variables in Regression Analysis](#)
- [Categorical Data Encoding Techniques](#)
- [Coding Schemes for Categorical Variables](#)

Building the regression equation: problem with a single coefficient

Previously: simple linear regression

- Outcome Y = numerical variable
- Predictor X = numerical variable

The regression (best-fit) line is:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

New: what if the explanatory variable is categorical?

Naively, we could write: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$

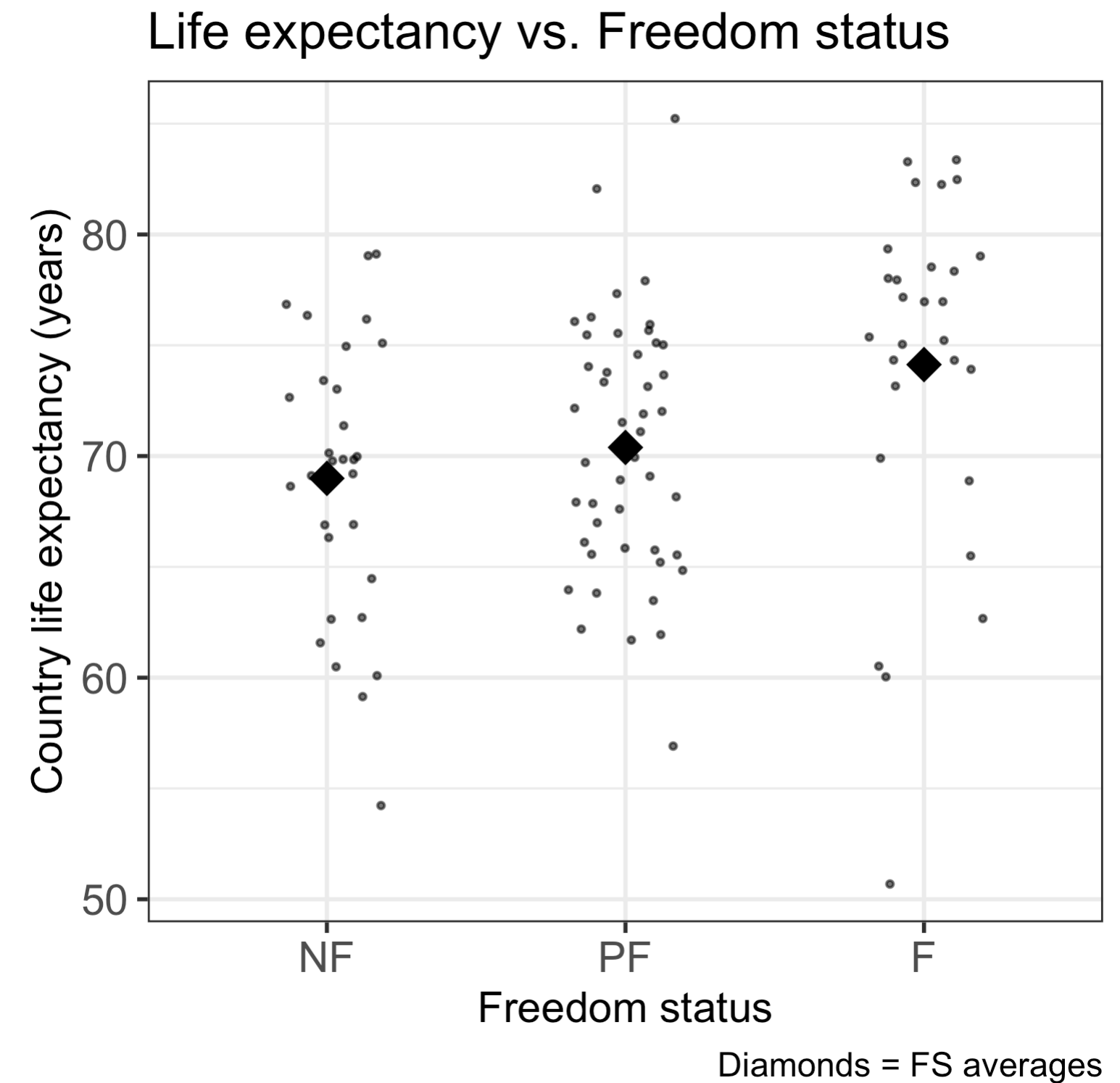
Or, with our variables:

$$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot FS$$

- But what does FS (freedom status) mean in this equation?
 - What values can it take? How do we represent each FS?
- Note: the above is WRONG

Building the regression equation: how do we map categories to means?

- If we only have freedom status in our model and want to map it to an average life expectancy...
- We want to create a function that can map each freedom status to life expectancy
 - If not free: $\widehat{LE} = 68.99$ years
 - If partly free: $\widehat{LE} = 70.39$ years
 - If free: $\widehat{LE} = 74.13$ years
- Can we make one equation for $\widehat{LE}$ by putting the “if” statements within the equation?



Building the regression equation: Indicator functions

- In order to represent each status in the equation, we need to introduce a new function:
 - Indicator function:

$$I(X = x) \text{ or } I(x) = \begin{cases} 1, & \text{if } X = x \\ 0, & \text{else} \end{cases}$$

- This basically a binary yes/no if X is a specific value x
- For example, if we want to identify a country as partly free, we can make:

$$I(FS = \text{partly free}) \text{ or } I(\text{partly free}) = \begin{cases} 1, & \text{if } FS = \text{partly free} \\ 0, & \text{else} \end{cases}$$

Poll Everywhere Question 1

Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables

2. Write the regression equation for a categorical variable using reference cell coding

3. Calculate and interpret coefficients for reference cell coding

4. Change the reference level in a categorical variable for reference cell coding

5. Create new variables and interpret coefficient for ordinal / scoring coding

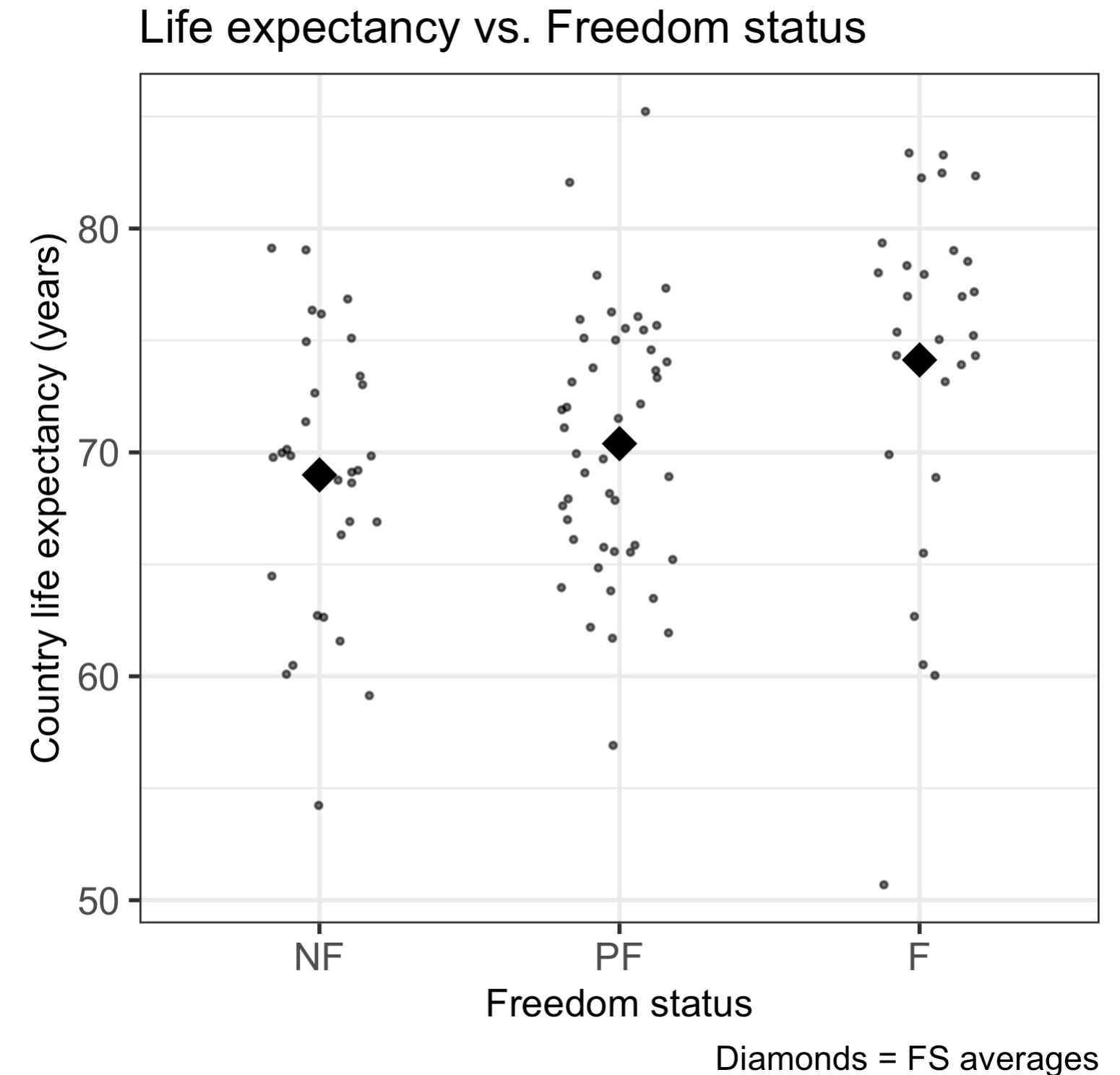
Building the regression equation: Indicators in our equation

$$\widehat{LE} = 68.99 \cdot I(\text{not free}) + 70.39 \cdot I(\text{partly free}) + 74.13 \cdot I(\text{free})$$

- However, a linear regression equation still requires an intercept!
 - So one of our statuses need to become our “reference” group
 - We’ll use “not free” as our reference
 - That means we need to adjust all the numbers

$$\widehat{LE} = 68.99 + 1.4 \cdot I(\text{partly free}) + 5.14 \cdot I(\text{free})$$

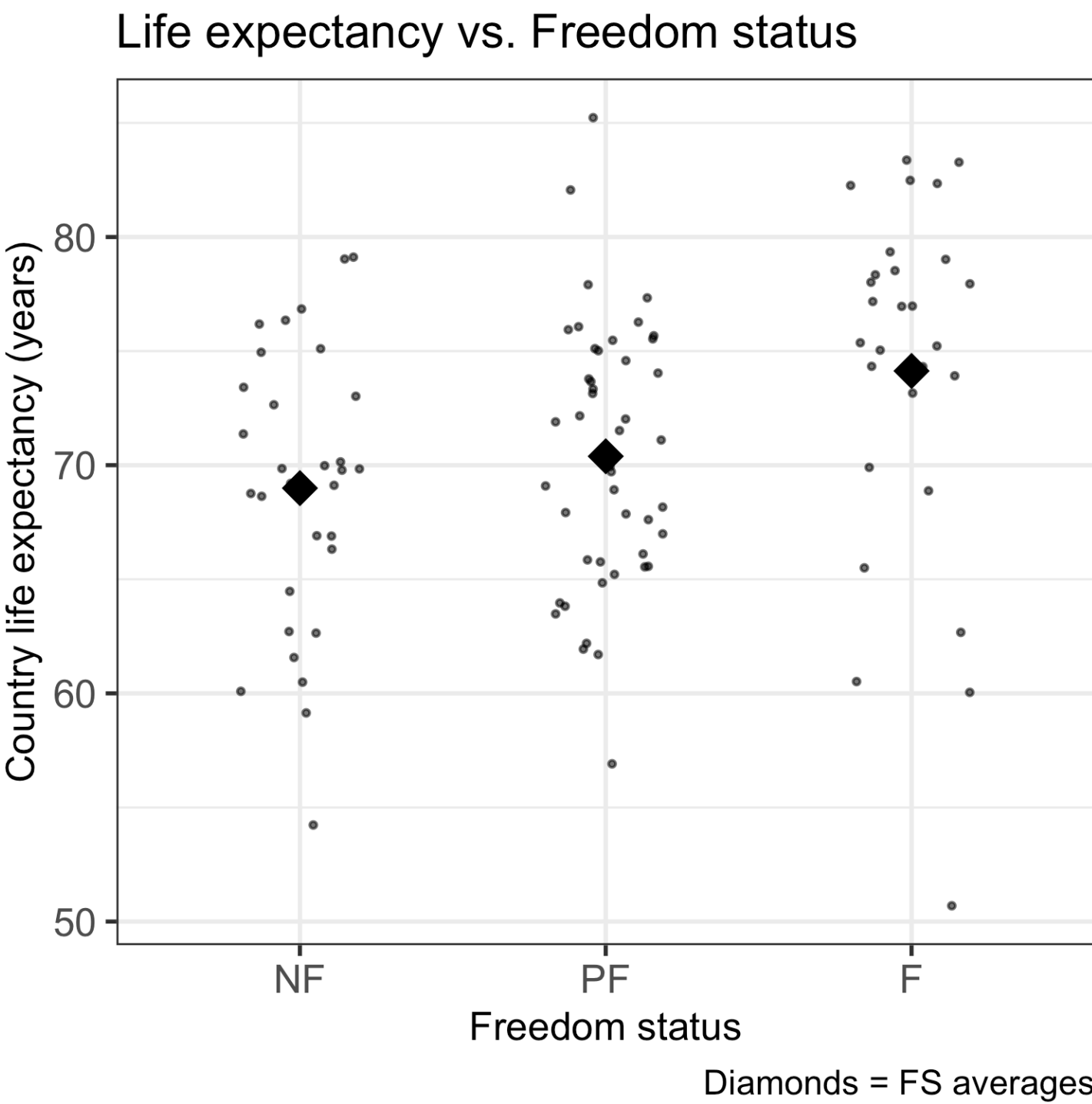
$$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{partly free}) + \hat{\beta}_2 \cdot I(\text{free})$$



Viewing the regression equation another way

$$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(FS = \text{partly free}) + \hat{\beta}_2 \cdot I(FS = \text{free})$$

Freedom status	Regression equation for FS	Average Life Expectancy for FS
Not free	$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 0$	$\widehat{LE} = \hat{\beta}_0$
Partly free	$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 0$	$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1$
Free	$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_1 \cdot 0 + \hat{\beta}_2 \cdot 1$	$\widehat{LE} = \hat{\beta}_0 + \hat{\beta}_2$



Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables
2. Write the regression equation for a categorical variable using reference cell coding
3. Calculate and interpret coefficients for reference cell coding
4. Change the reference level in a categorical variable for reference cell coding
5. Create new variables and interpret coefficient for ordinal / scoring coding

Interpretation of regression equation coefficients

Remember: expected, mean, and average are interchangeable

Coefficient	Interpretation
	Expected/mean/average life expectancy of countries that are not free
	Difference in mean life expectancy between partly free and not free countries - OR- Mean difference in life expectancy between partly free and not free countries
	Difference in mean life expectancy between free and not free countries -OR- Mean difference in life expectancy between free and not free countries

Poll Everywhere Question 2

Regression table with `lm()` function

```
1 model1 = gapm %>%  
2   lm(formula = life_exp ~ freedom_status)  
3  
4 tidy(model1, conf.int=T) %>% gt() %>% tab_options(table.font.size = 38) %>%  
5   fmt_number(decimals = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	68.99	1.17	58.88	0.00	66.67	71.32
freedom_statusPF	1.40	1.52	0.92	0.36	-1.61	4.40
freedom_statusF	5.14	1.70	3.02	0.00	1.76	8.51

$$\widehat{LE} = 68.99 + 1.4 \cdot I(PF) + 5.14 \cdot I(F)$$

- Which Freedom status did R choose as the reference level?
- How you would calculate the mean life expectancies of freedom statuss using *only* the results from the regression table?

Bringing in the numbers/units/95% CI

Coefficient	Interpretation
$\hat{\beta}_0$	Average life expectancy of countries that are not free is 68.99 (95% CI: 66.67, 71.32).
$\hat{\beta}_1$	The difference in mean life expectancy between countries that are partly free and not free is 1.4 (95% CI: -1.61, 4.4).
$\hat{\beta}_2$	The difference in mean life expectancy between countries that are free and not free is 5.14 (95% CI: 1.76, 8.51).

- Don't forget that we can use the confidence intervals to assess whether the mean difference with "not free" is significant or not

We can also use R to report each status's average life expectancy

Find the 95% CI's for the mean life expectancy for the countries that are partly free and free

- Use the base R `predict()` function (see Lesson 4 for more info)
- Requires specification of a `newdata` "value"

```
1 newdata = data.frame(freedom_status = c("NF", "PF", "F"))
```

```
1 (pred = predict(model1,  
2           newdata=newdata,  
3           interval="confidence"))
```

	fit	lwr	upr
1	68.99387	66.66958	71.31816
2	70.39000	68.48194	72.29806
3	74.12893	71.68329	76.57457

Interpretations

- The average life expectancy for countries that are partly free is 70.39 years (95% CI: 68.48, 72.3).
- The average life expectancy for countries that are free is 74.13 years (95% CI: 71.68, 76.57).

Another way to look at coefficient values

$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{PF}) + \hat{\beta}_2 \cdot I(\text{F})$$

► Code

Freedom status	Average life expectancy	Difference with NF
NF	69.0	0.0
PF	70.4	1.4
F	74.1	5.1

$$\widehat{\text{LE}} = 68.99 + 1.4 \cdot I(\text{PF}) + 5.14 \cdot I(\text{F})$$

10 minute break here?

Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables
2. Write the regression equation for a categorical variable using reference cell coding
3. Calculate and interpret coefficients for reference cell coding
4. Change the reference level in a categorical variable for reference cell coding
5. Create new variables and interpret coefficient for ordinal / scoring coding

Reference levels

Why is **NF** not one of the variables in the regression equation?

$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{PF}) + \hat{\beta}_2 \cdot I(\text{F})$$

- Categorical variables must have at least 2 levels. If they have 2 levels, we call them *binary*
- We choose one level as our **reference level** to which all other levels of the categorical variable are compared
 - The levels and \$ are compared to the level not free
- The **intercept** of the regression equation is the *mean of the outcome restricted to the reference level*
 - Recall that the intercept is the mean life expectancy of countries that are not free, which was our reference level
- If the categorical variable has r levels, then we need $r - 1$ variables/coefficients to model it!

We can change the reference level to PF (1/2)

- Suppose we want to compare the mean life expectancies of freedom statuses to the partly free level instead of not free
- Below is the estimated regression equation for when not free is the reference level

$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{PF}) + \hat{\beta}_2 \cdot I(\text{F})$$

- Update the variables to make partly free the reference level:

$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{NF}) + \hat{\beta}_2 \cdot I(\text{F})$$

We can change the reference level to PF (2/2)

- Now update the coefficients of the regression equation using the output below.

Freedom status	Average life expectancy	Difference with PF
NF	68.99	-1.40
PF	70.39	0.00
F	74.13	3.74

$$\widehat{\text{LE}} = 70.39 - 1.4 \cdot I(\text{NF}) + 3.74 \cdot I(\text{F})$$

R: Change reference level to PF

- `freedom_status` data type was originally a `character`
- check this with `str()` or `class()` or `glimpse()`

```
1 str(gapm$freedom_status)
```

Factor w/ 3 levels "NF","PF","F": 1 2 1 3 2 1 2 2 3 2 ...

- To change the reference level, we can use `fct_relevel()` from the `forcats` package (part of `tidyverse`)
- *Any levels not mentioned will be left in their existing order, after the explicitly mentioned levels.*

```
1 gapm2 = gapm %>%  
2   mutate(  
3     freedom_status = fct_relevel(freedom_status, "PF")  
4   )
```

- Check the order:

```
1 levels(gapm2$freedom_status)
```

```
[1] "PF" "NF" "F"
```

R: Run model with PF as the reference level

```
1 levels(gapm2$freedom_status)
```

```
[1] "PF" "NF" "F"
```

```
1 model2 = gapm2 %>% lm(formula = life_exp ~ freedom_status)
2 tidy(model2) %>% gt() %>% tab_options(table.font.size = 35) %>%
3   fmt_number(decimals = 2)
```

term	estimate	std.error	statistic	p.value
(Intercept)	70.39	0.96	73.17	0.00
freedom_statusNF	-1.40	1.52	-0.92	0.36
freedom_statusF	3.74	1.56	2.39	0.02

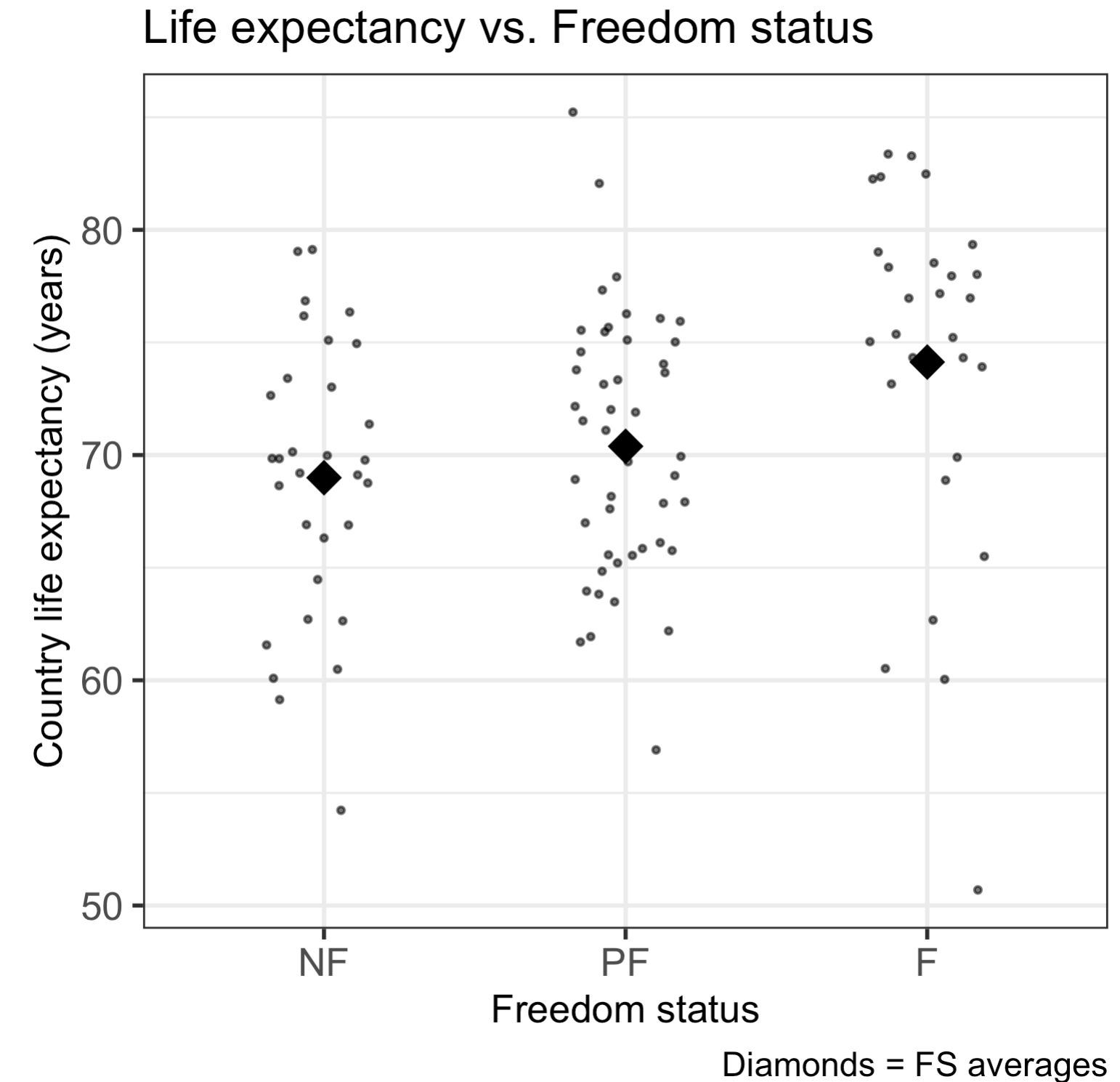
$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot I(\text{NF}) + \hat{\beta}_2 \cdot I(\text{F}) +$$

$$\widehat{\text{LE}} = 70.39 - 1.4 \cdot I(\text{NF}) + 3.74 \cdot I(\text{F})$$

Fitted values & residuals

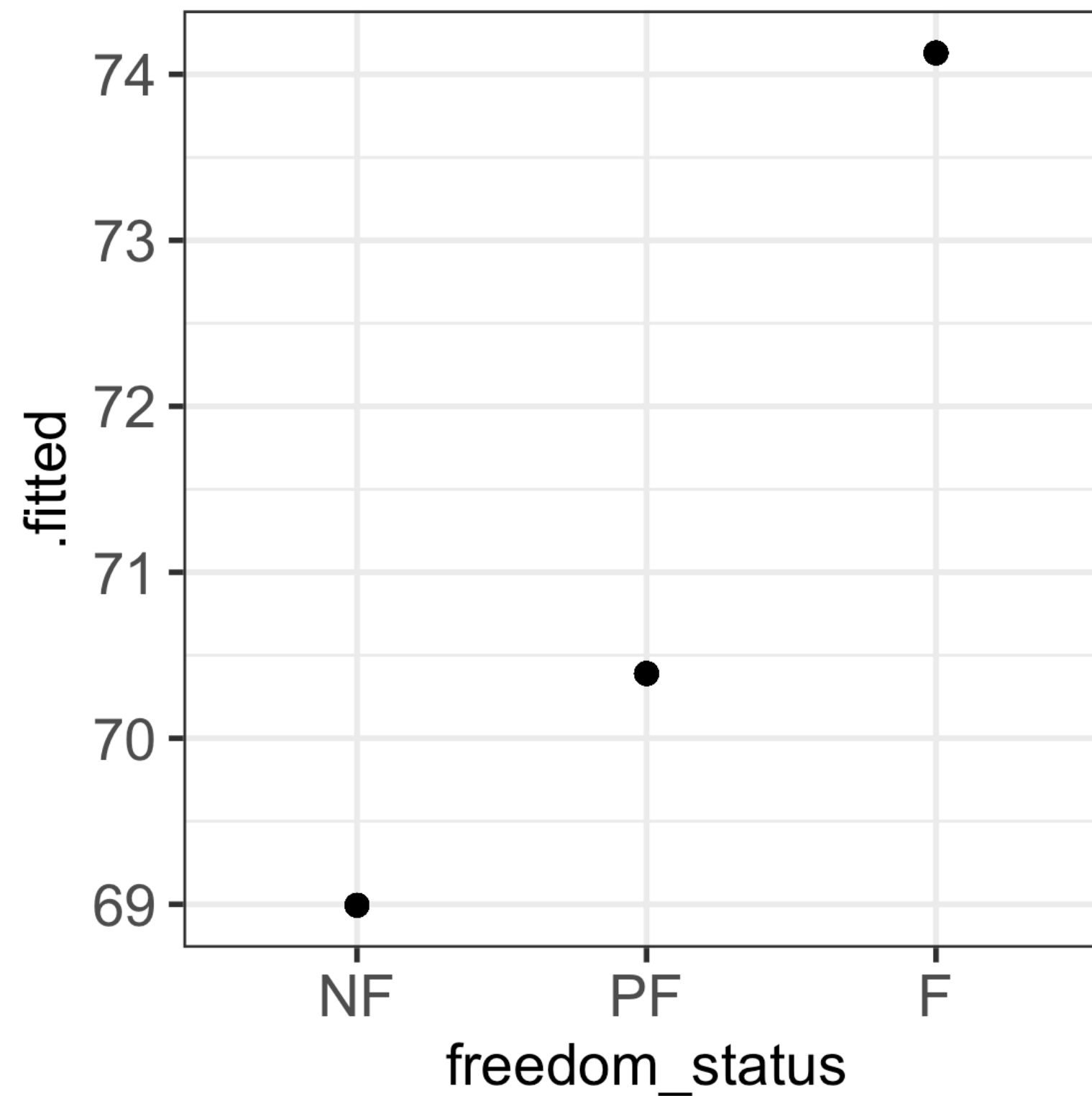
Similar to as before:

- **Observed values** Y are the values in the dataset
- **Fitted values** $\hat{Y}$ are the values that ~~fall on the best-fit line for a specific value of x~~ are the *means of the outcome stratified by the categorical predictor's levels*
- **Residuals** ($\hat{\epsilon} = Y - \hat{Y}$) are the differences between the two



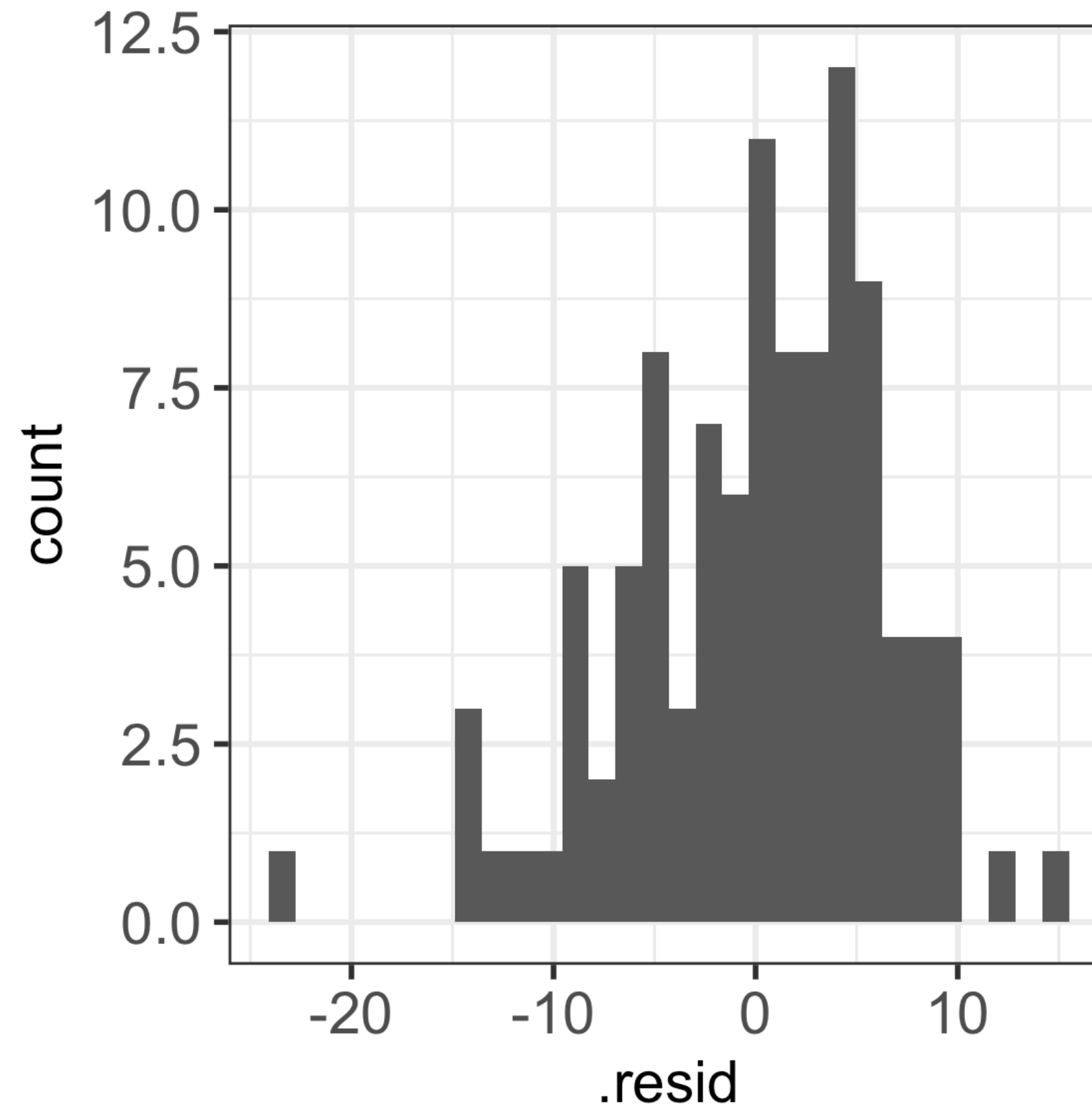
Fitted values are the same as the means

```
1 m1_aug <- augment(model1)
2
3 ggplot(m1_aug, aes(x = freedom_status, y = .fitted)) + geom_point() +
4   theme(axis.text = element_text(size = 22), axis.title = element_text(size = 22))
```



Residual plots (now the spread within each status)

```
1 ggplot(m1_aug, aes(x=.resid)) + geom_histogram() +  
2   theme(axis.text = element_text(size = 22), title = element_text(size = 22))
```



Poll Everywhere Question 3

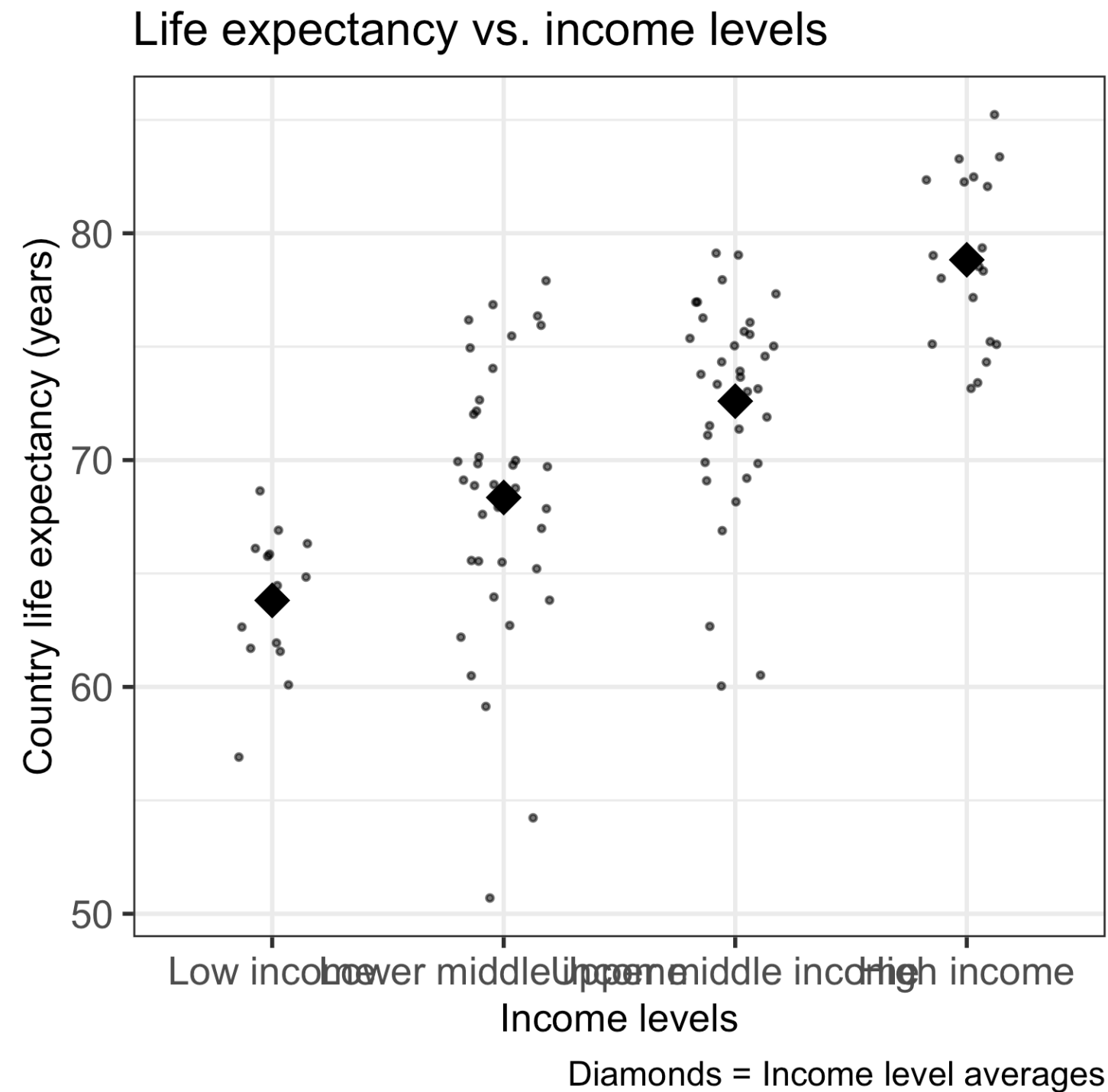
Learning Objectives

1. Understand why we need a new way to code categorical variables compared to continuous variables
2. Write the regression equation for a categorical variable using reference cell coding
3. Calculate and interpret coefficients for reference cell coding
4. Change the reference level in a categorical variable for reference cell coding
5. Create new variables and interpret coefficient for ordinal / scoring coding

Let's look at life expectancy vs. four income levels

- Gapminder discusses individual income levels
- **Income levels for a country** is based on average GDP per capita, and grouped into:
 - Low income
 - Lower middle income
 - Upper middle income
 - High income

Visualizing the ordinal variable, income levels



A few changes needed:

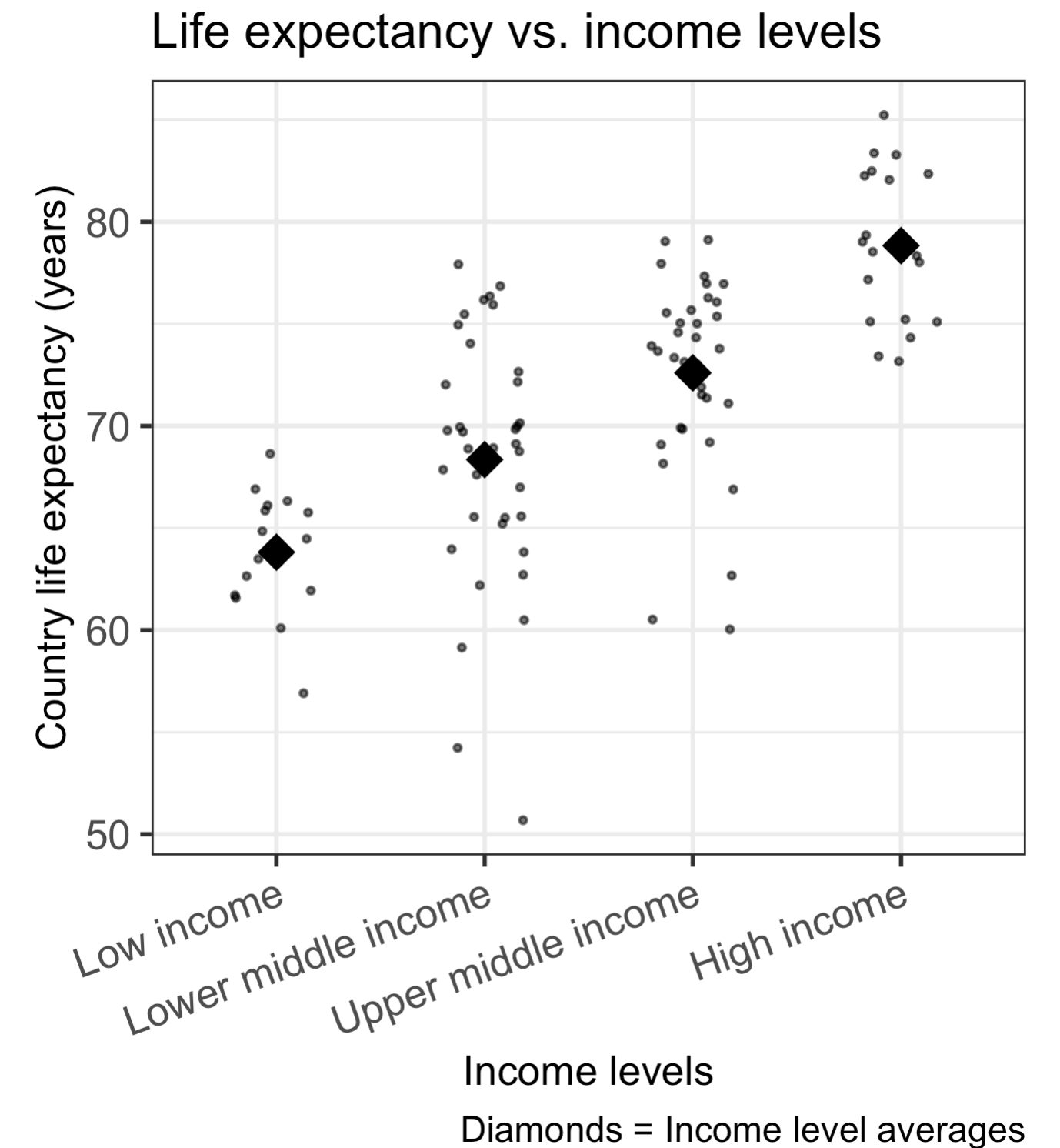
- Put the income levels in order

```
1 gapm2 = gapm2 %>%  
2   mutate(income_level_4 = fct_relevel(  
3     income_level_4,  
4     "Low income",  
5     "Lower middle income",  
6     "Upper middle income",  
7     "High income"  
8   ))
```

- Make the income levels readable
 - **How to Rotate Axis Labels in ggplot2?**

Much better: Visualizing the ordinal variable, income levels

```
1 ggplot(gapm2, aes(x = income_level_4, y = life_exp)) +  
2   geom_jitter(size = 1, alpha = .6, width = 0.2) +  
3   stat_summary(fun = mean, geom = "point", size = 8, shape = 18) +  
4   labs(x = "Income levels",  
5        y = "Country life expectancy (years)",  
6        title = "Life expectancy vs. income levels",  
7        caption = "Diamonds = Income level averages") +  
8   theme(axis.title = element_text(size = 20),  
9         axis.text = element_text(size = 20),  
10        title = element_text(size = 20),  
11        axis.text.x=element_text(angle = 20, vjust = 1, hjust=1))
```



How can we code this variable?

We have two options:

Treat the levels as nominal, and use reference cell coding

- Like we did with freedom status
- This option will not break the linearity assumption
- For g categories of the variable, we will have $g - 1$ coefficients to estimate

Use the ordinal values to score the levels and treat as a numerical variable

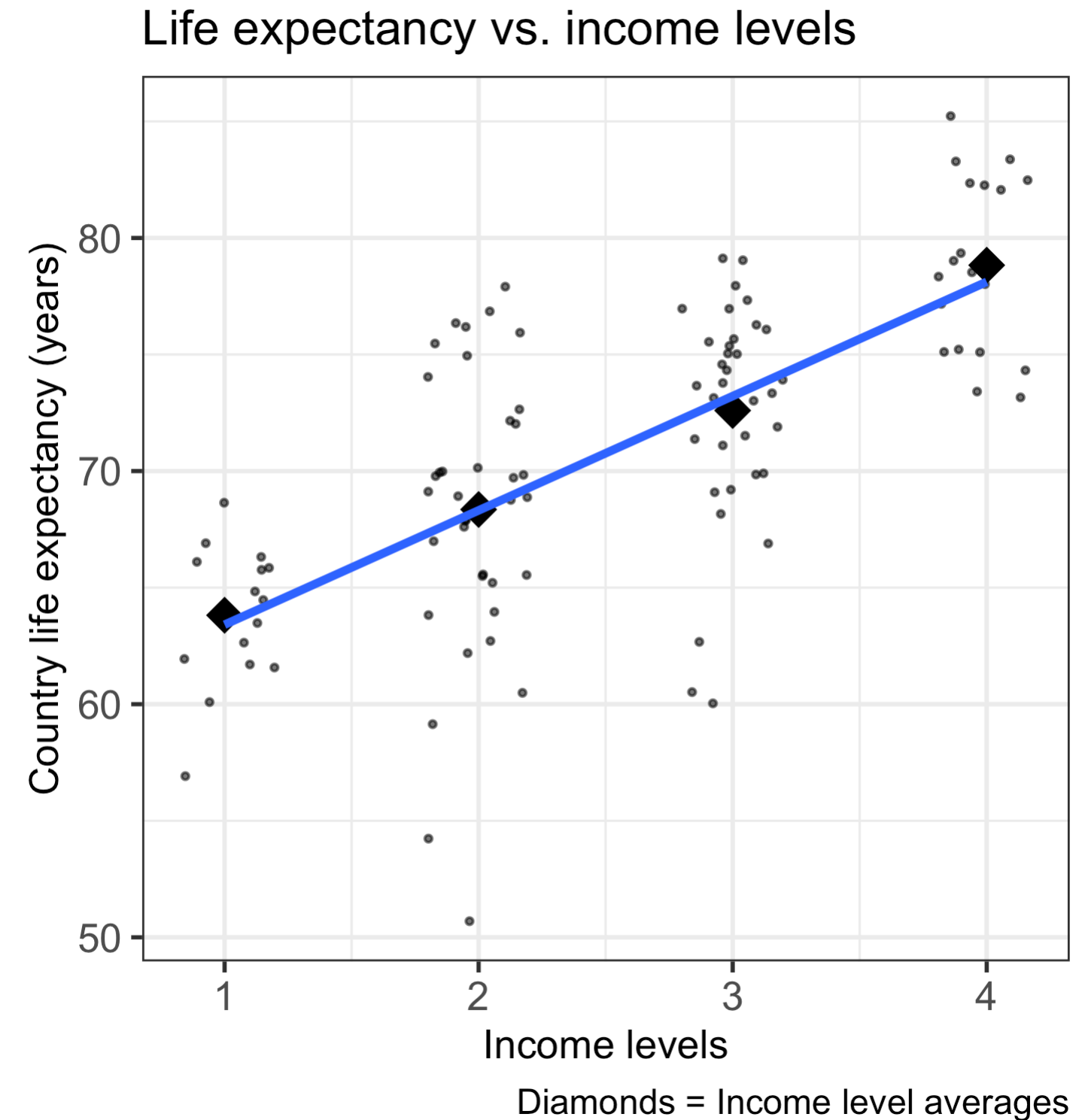
- Even if a variable is inherently ordered, we need to check that linearity holds if categories are represented as numbers
- This way of coding preserves more power in the model (less coefficients to estimate means more power)
- Only one coefficient to estimate

Some important considerations when scoring ordinal variables

- Even if a variable is inherently ordered, we need to check that linearity holds if categories are represented as numbers (more in next lessons)
 - Linearity is an assumption of linear regression: that the relationship between X and Y is linear
- Assumes differences between adjacent groups are equal
 - Income levels are pre-set groups by Gapminder
 - Might be hard to interpret “every 1-level increase in income level”
- Is the variable part of the main relationship that you are investigating? (even if linearity holds)
 - If yes, consider leaving as reference cell coding unless the interpretation makes sense
 - If no, and just needed as an adjustment in your model, then power benefit of scoring might be worth it!

Check that linearity holds for income levels

- Using visual assessment, linearity holds for our income levels (more in next lessons)
- We can use the ordinal encoding for income levels



Poll Everywhere Question 4

Ordinal coding / Scoring

- Map each income level to a number
- Usually start at 1

Income Level	Score
Low income	1
Lower middle income	2
Upper middle income	3
High income	4

```
1 gapm2 = gapm2 %>%
2   mutate(income_num = case_when(
3     income_level_4 == "Low income" ~ 1,
4     income_level_4 == "Lower middle income" ~ 2,
5     income_level_4 == "Upper middle income" ~ 3,
6     income_level_4 == "High income" ~ 4
7   ))
8 gapm2 %>% select(income_level_4, income_num) %>%
9   head(6)
```

A tibble: 6 × 2

income_level_4	income_num
<fct>	<dbl>
1 Low income	1
2 Upper middle income	3
3 High income	4
4 Upper middle income	3
5 Upper middle income	3
6 Upper middle income	3

Run the model with the scored income

```
1 mod_inc2 = gapm2 %>% lm(formula = life_exp ~ income_num)
2 tidy(mod_inc2) %>% gt() %>% tab_options(table.font.size = 37) %>%
3   fmt_number(decimals = 2)
```

term	estimate	std.error	statistic	p.value
(Intercept)	58.51	1.38	42.47	0.00
income_num	4.90	0.51	9.66	0.00

$$\widehat{\text{LE}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Income level}$$

$$\widehat{\text{LE}} = 58.51 + 4.9 \cdot \text{Income level}$$

- Keep in mind: We cannot calculate the expected outcome outside of the scoring values
 - For example, we cannot find the mean life expectancy for an income level of 1.5

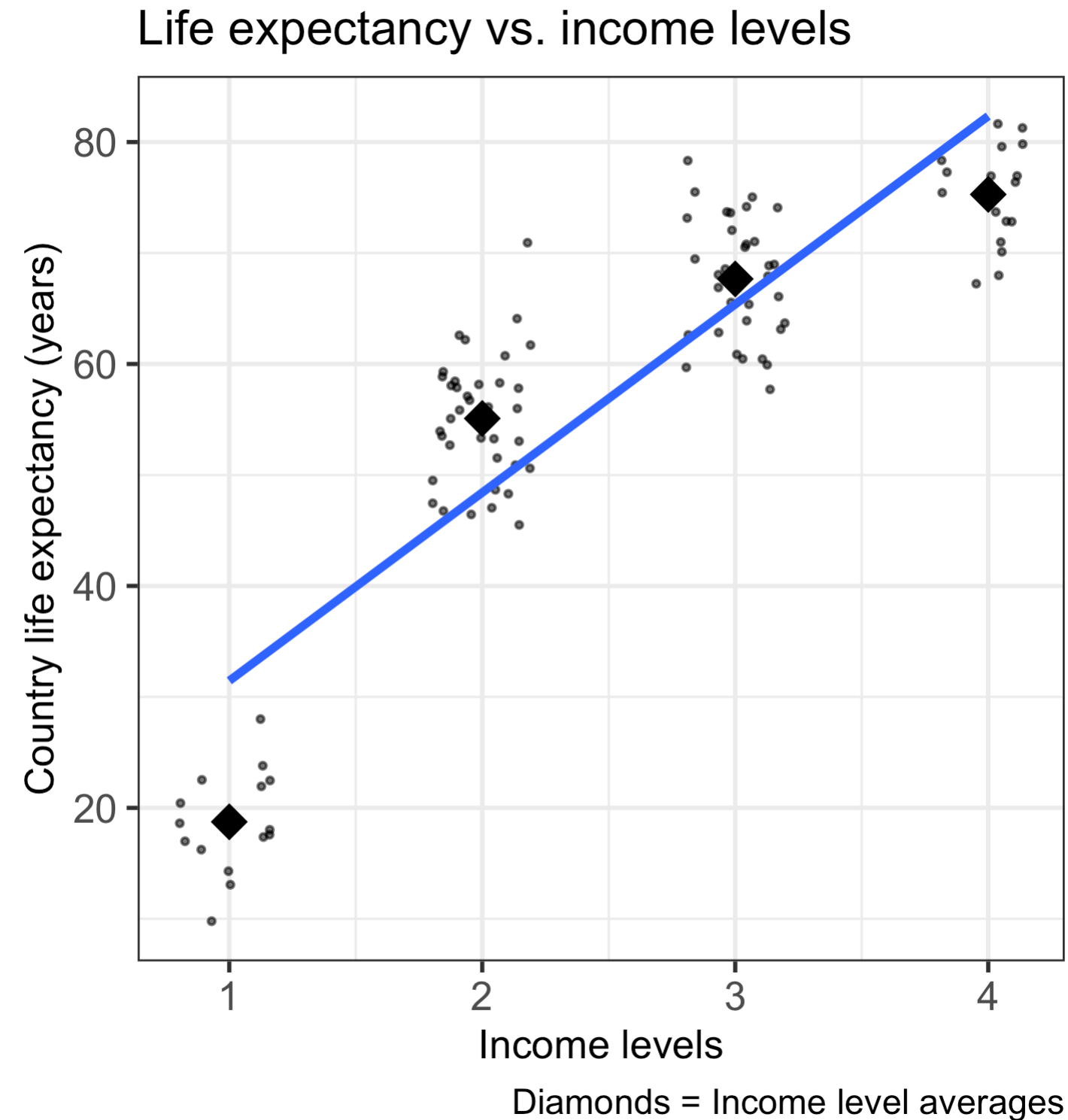
Interpreting the model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	58.51	1.38	42.47	0.00	55.78	61.24
income_num	4.90	0.51	9.66	0.00	3.90	5.91

$$\widehat{\text{LE}} = 54.01 + 6.25 \cdot \text{Income level}$$

- **Interpreting the intercept:** At an income level of 0, mean life expectancy is 54.01 (95% CI: 51.92, 56.10).
 - Note: this does not make sense because there is no income level of 0!
- **Interpreting the coefficient for income:** For every 1-level increase in income level, mean life expectancy increases 4.9 years (95% CI: 3.9, 5.91).

What if life expectancy vs. income level looked like this?



- No longer maintaining the linearity assumption
- Need to use reference cell coding
- We would fit the following model:

$$\text{LE} = \beta_0 + \beta_1 \cdot I(\text{Lower middle income}) + \beta_2 \cdot I(\text{Upper middle income}) + \beta_3 \cdot I(\text{High income}) + \epsilon$$

If time...

Let's walk through categorical variables that have multiple selections

- So each group is not mutually exclusive
- We could make an indicator for each category, but individuals could be a part of multiple categories
- Also, thinking about income levels - can we combine two groups to make one??